

Audit Exception Engine for Bank Statement Analytics

Problem Statement

During statutory audits, internal audits, tax audits, and financial reviews, auditors often need to examine a large volume of bank transactions to identify unusual, duplicate, high-value, or potentially risky transactions.

Manual review of bank statements is time-consuming and highly dependent on the auditor's ability to identify patterns across hundreds or thousands of transactions. Important exceptions may be overlooked due to the volume of data, repetitive transaction descriptions, different narration formats, and the absence of automated analytical procedures.

The proposed **Audit Exception Engine for Bank Statement Analytics** is designed to automate the preliminary analysis of bank statements and assist Chartered Accountants and audit teams in identifying transactions that require further investigation.

The system will accept a bank statement in Excel or CSV format containing transaction date, narration, cheque/reference number, debit amount, credit amount, and closing balance. The system will normalize the data, classify transactions, apply predefined audit rules, identify exceptions, assign risk levels, and generate an audit-oriented exception report.

Key Audit Problems Addressed

The system will automatically identify:

1. Large Transactions

2. Transactions exceeding a configurable threshold.
3. Separate analysis of large debit and credit transactions.

4. Large Cash Transactions

5. Significant cash deposits.
6. Significant cash withdrawals.
7. Identification of repeated large cash transactions.

8. Repeated Transactions

9. Repeated transactions with similar narration, amount, date, or reference information.
10. Identification of potentially duplicate or repetitive payments.

11. Duplicate References

12. Repeated cheque/reference numbers.

13. Potential duplicate transaction references requiring verification.

14. Round-Value Transactions

15. Transactions involving significant round amounts such as ₹1,00,000, ₹5,00,000, etc.

16. Identification of patterns of repeated round-value transactions.

17. Unusual Narration

18. Identification of narrations containing unusual, unknown, manual, adjustment, or unidentified descriptions.

19. Identification of transactions that do not fit normal transaction patterns.

20. Unusual Transaction Frequency

21. Identification of unusually high transaction frequency for a particular party, narration, or transaction category.

22. Detection of sudden bursts of transactions.

23. Sudden Changes in Transaction Values

24. Comparison of transaction values over time.

25. Identification of significant increases or decreases compared with historical transaction patterns.

26. Unknown Parties

27. Identification of parties that appear only once or cannot be classified.

28. Identification of transactions involving previously unseen parties/narrations.

29. Transfers Between Accounts

- Identification of NEFT, RTGS, IMPS, UPI, self-transfer, and other fund-transfer transactions.
- Identification of significant or repeated transfers requiring audit review.

30. Investment-Related Transactions

- Identification of transactions related to mutual funds, brokerage accounts, securities, investment platforms, etc.
- Separate reporting of investment-related debits and credits.

Objective

The primary objective is to create an automated **first-level audit screening mechanism** that reduces manual effort and helps auditors focus their attention on transactions having higher audit significance.

The system will not determine that a transaction is fraudulent or incorrect. Instead, it will identify transactions that meet predefined analytical criteria and present them as **audit exceptions requiring professional review**.

Expected Output

The system will generate an exception report containing:

- Transaction Date
- Narration
- Cheque/Reference Number
- Debit
- Credit
- Transaction Category
- Exception Type
- Exception Reason
- Risk Level
- Rule Triggered
- Suggested Audit Procedure

The system will also provide a dashboard showing:

- Total transactions reviewed
- Total debit value
- Total credit value
- Number of exceptions
- High-risk exceptions
- Medium-risk exceptions
- Large cash transactions
- Large-value transactions
- Repeated transactions
- Investment-related transactions
- Monthly transaction trends

This system will provide auditors with a structured, transparent, and repeatable approach to bank statement analytics while retaining the requirement for professional judgement by the Chartered Accountant.